



LTV REPAIR PROCEDURES

Relevant to: EV Teams

NASA SUITS
2025-26 Challenge

This document is mostly finalized for Test Week 2026, but still subject to change

Contents

Notes	1
Change Log:	2
Error Format & Interface	2
Example Errors	3
Interpreting & Triaging Errors	3
Error Interface	3
Procedures	3
Exit Recovery Mode (4800).....	4
NAV Restart & Manual Return to Home (4509)	5
Backup Fuse Error (1969)	6
Poor Comms RSSI (3452)	6
Comms Reboot Required (2441)	7
Subsystem Power Bus Error (4968)	8
Dust Sensor Error (2235)	8
Scientific Power Distribution Error (4280)	8

Notes:

NASA SUITS aims to have a physical activity box with various panels and mock sensors for the EV to troubleshoot and repair. This physical box is representative of the Lunar Terrain Vehicle (LTV) External Control Panel and will be referred to as the “LTV Task Board”. Each item within the LTV Task Board will have a unique procedure for troubleshooting and repair, this document contains each of those procedures.

These procedures are to be retrieved by your system from the TSS and during Test Week, your assigned Design Evaluator will be the one performing the procedures. **Do NOT store these procedures**, only reference them for understanding how the challenge will proceed. A complete procedure list will be available in the TSS and shall be retrieved by your Artificial Intelligence Assistant (AIA) during Test Week.

Change Log:

- 1. Flesh out initial tentative LTV Repair procedures, add cover page, general formatting, write *Notes* and *Procedures* section, override example errors with planned errors -RE 2/17/2026
- 2. Combine RTH and NAV Restart into one procedure, make comms error relevant to physical box, rewrite ERM to be a prerequisite to anything else, describe error formatting, error notes, and provide examples -RE 3/3/2026
- 3. Reiterate all current errors to follow format and layout of current task board, refine language and steps for errors that were previously vague -RE 4/9/2026
- 4. Unify language and implement final variations of errors relevant to final task board. Specify more fore error interface -RE 5/11/2026

Error Format & Interface

Each procedure will have an error code associated with it. These error codes will be physically displayed on a seven-segment display on the LTV Task Board as well as existing in the TSS. These errors will consist of 4 digits corresponding to the following format:

Criticality Class	Subsystem Identifier	Error Identifier	Error Identifier
1-4	0-9	0-9	0-9

The subsystem identifier digit corresponds to the following list:

ID	Subsystem
0	Pre-Flight Integration and Testing
1	Secondary Computer Human Interface
2	Scientific Instrumentation
3	Lighting
4	Communications
5	Autonomous Guidance & Navigation
6	Vehicular Mobility & Control
7	Thermal Regulation
8	Diagnostic & System Monitoring
9	Power Distribution

Example Errors

Error 4965: A high criticality error within the Power Subsystem, identifier 65

Error 3224: A moderate-to-high criticality error within the Scientific Instrumentation Subsystem, identifier 24

Error 2777: A moderate criticality error within the Thermal Regulation Subsystem, identifier 77

Error 1401: A very low criticality error within the Lighting Subsystem, identifier 01

Interpreting & Triageing Errors

- Both the criticality class and the subsystem identifier follow a lowest-to-highest priority format, where 0/1 is the lowest priority, and conversely, 9 is the highest priority.
 - The error identifiers are **NOT** indicative of any priority; they serve solely to distinguish different errors.
- The criticality class corresponds to the subsystem, meaning two errors with equal criticality class should be prioritized based on which subsystem they correspond to.
 - Ex: Error 2700 should be fixed before 2600, and Error 3900 should be fixed before Error 2900
 - Teams should use their AIA to determine this ordering, but will also have to triage large disparities, such as choosing which to fix between a higher critically on a lower priority subsystem or a lower criticality on a higher priority subsystem (Ex: Error 4200 vs. Error 1800)
 - In some scenarios, it may be better to fix the higher criticality error on the lower priority subsystem
 - Teams are encouraged to use their AIA to triage errors in a non-linear fashion, considering exactly which subsystems are having the error and how those might affect the overall function of the LTV

Error Interface

These error codes will also exist in the TSS as a JSON file, named `LTV\_ERROS.json` which will contain Booleans and the relevant procedures to fix these errors, please check the TSS documentation to reference the format of this JSON. This is to allow your system to assess which errors are present and simultaneously access the procedures to fix them.

It is imperative to continually check for errors as new errors may occur during your test.

Procedures

The primary task during the test session for EV Teams will be to repair the LTV systems that were damaged during the loss of communication. The following procedures will detail how to Exit Recovery Mode (ERM), Restart the navigation (NAV) system, as well as repair the LTV for each of the errors that could be indicated from the TSS. These errors will vary in criticality, and as per the [25-26 Mission](#)

Description, some fixes may be omitted if the time remaining is low. Subsequently, your AIA may be able to triage which errors to attempt to fix first. Additionally, as these procedures are iterated, they will become very verbose, and your AIA may shorten the steps in the procedures if able to do so while keeping critical information pertinent to the task.

Do NOT store these procedures, only reference them for understanding how the challenge will proceed. Procedures should be sourced from the TSS as mentioned previously.

Exit Recovery Mode (4800)

1. Investigate for prominent physical damage to any components before proceeding. This may include wires with damaged insulation, dents or large cracks visible on components, or missing wires
2. Locate the power distribution and diagnostic (PDD) component by identifying an irregular-shaped white housing labeled with “PDD9900xxxx”
3. On the PDD display, verify the vehicle is in recovery mode and not some other error state, this is indicated by the error code 4800 and a red light by the recovery switch, labeled “RECO”
 - a. If a different error is displayed, that error must be fixed prior to exiting recovery mode
4. Via the Power Override Panel for Subsystems, labeled “POPS”, move switches for “MPS”, “GNC”, and “CDH” to the “OFF” position
5. Locate the Software Fault Acknowledge button, labeled “SFA”
6. Press and release the SFA button
7. Move the Power Optimization Mode switch, labeled “POM”, to the “OFF” position, verify change by confirming a blue indicator light
8. Open the switch cover for the External Battery Disconnect switch, labeled “EBD”
9. Press and hold the Computer Software Reboot button, labeled “CSR”
10. While holding the CSR button, move the EBD switch to the “ON” position
11. Release the CSR button
12. Wait 5-10 seconds for the reboot to take effect. Successful reboot will be indicated by the green light on the software management box turning on
13. Move the EBD switch to the “OFF” position and close the switch cover
14. Identify the Software Management and Control (SMC) component by identifying a rectangular white housing labeled with “SMC1100xxxx”
15. Verify the red status light on the SMC begins blinking
16. Via POPS, move the switches for MPS, GNC, and CDH to the “ON” position
17. Open the switch cover for the control switch, labeled “CTRL”, on the SMC
18. On the SMC, press and hold the black alternative button, labeled “ALT”, while moving the CTRL switch to the on position
19. Verify the red status light on the SMC turns off
20. Release the ALT button
21. Test the reboot by pushing the “TEST” switch upward on the SMC and pressing and holding the CSR button on the PDD
22. Verify the green status light on the SMC has begun blinking

23. On the SMC, press and hold the black ALT button while switching off the CTRL switch to the “OFF” position
24. Verify a solid green status light has illuminated
25. Close the switch cover of the CTRL switch
26. On the PDD, open the switch cover for the recovery switch, labeled “RECO”
27. Move the RECO switch to the “OFF” position and close the cover
28. Verify the LTV has exited recovery mode by confirming the error code 4800 no longer displays and the RECO indicator light is no longer illuminated
29. Announce ERM success

NAV Restart & Manual Return to Home (4509)

1. Locate the Navigation Control Panel (NAV) by identifying a rectangular white housing labeled “NAV5500xxxx”
2. Move the primary control switch under the green light to the far left, entering “HAND” mode. Verify mode switch by confirming the mode indicator light turns blue
3. Identify the “LIDAR RESET” button
4. Flush and restart the LIDAR system by pressing and holding this button for 5 seconds. The LIDAR button indicator will light red upon depressing this button
5. On the SMC, verify the green status light has illuminated, indicating a successful LIDAR system restart
6. On the NAV component, Identify the “NAV RESET” button
7. Flush and restart the NAV System by pressing and holding this button for 5 seconds. The NAV button indicator will light red upon depressing this button
8. On the SMC, verify the red indicator light has illuminated, indicating a successful NAV system restart
9. On the NAV component, enact the reset by moving the primary control switch to the “OFF” position and wait for 5 seconds
10. Move the primary control switch to HAND mode
11. Verify mode switch by confirming the mode indicator light is blue
12. Identify the autonomous systems indicator toggle switch labeled “ASITS”
13. Move the ASITS switch to the “ON” position
14. Verify mode switch by confirming the indicator light is yellow and the remaining indicators are on
15. Identify the autonomous navigation system lock labeled “ANAV BLOCK”
16. Move this switch to the “ON” position, verify mode switch by confirming the indicator light is blue
17. Identify the manual return to home switch labeled “ANAV RTH”
18. Move this switch to the “ON” position, verify mode switch by confirming the indicator light is blue
19. Identify the autonomy confidence adjustment dial labeled “ACA”
20. Rotate this dial counterclockwise to manually reduce the confidence scale of the autonomous navigation algorithm until the light is red
21. Move the ANAV BLOCK switch to the “OFF” position
22. Move the ASITS switch to the “OFF” position

23. Identify the autonomous Cognitive Obstacle and Motion Manager switch, labeled “COMM”
24. Turn the switch to the “SEC” position, and verify mode switch by confirming the indicator light is blue
25. On the SMC, verify the status lights have turned off
26. Move the primary control switch to the “AUTO” position, verify mode switch by confirming mode indicator light is green
27. Announce successful NAV Restart & Manual Return to Home steps complete

Backup Fuse Error (1969)

1. Locate the Backup Fuse Housing by identifying a rectangular gray housing labeled “BFH9900xxxx”
2. Lift the lid of the enclosure, exposing a black protective barrier, ensure the lid locks in the upright position
3. Before proceeding, be cognizant of sharp edges, pinch points, and where necessary, hold the enclosure steady with one hand while performing the following steps
4. Remove the fuse disconnect by pulling the exposed black handle located towards the top of the box, moderate force may be required
5. Remove the protective barrier by pulling upward from the opening marked by the “PULL” label
6. Use the provided fuse-puller tool to safely remove both fuses and properly discard them
7. Locate the replacement fuses
8. Using the fuse-puller tool, place a new fuse in the grips of the tool
9. Insert the fuse into the empty slot, moderate force may be required
10. Repeat steps 7 and 8 for the other fuse
11. Replace the black protective barrier
12. Reinsert the fuse disconnect in the proper orientation
13. Close the lid of the enclosure by lifting slightly upwards and pulling down
14. Announce Backup Fuse Error resolved

Poor Comms RSSI (3452)

1. Locate the antenna calibration and modification (ACM) component by identifying a square gray housing labeled “ACM4400xxxx”
2. Locate the phase control knob, labeled “PHS”, rotate this until maximum possible RSSI improvement is achieved
3. Locate the modulation control knob, labeled “MOD”, rotate until maximum possible RSSI improvement is achieved
4. Locate the amplitude control knob, labeled “AMP”, rotate until maximum possible RSSI improvement is achieved
5. If the green indicator has not been lit, you may rotate these controls until maximum RSSI strength has been achieved, this is indicated by the green indicator being lit
6. On the SMC component, verify the red status light has begun blinking
7. On the SMC, open the switch cover of the CTRL switch
8. Press and hold the black ALT button while moving the CTRL switch to the “ON” position
9. Verify the red status light is no longer illuminated

10. To finish resetting the software for the ACM component, press and hold the black ALT button and the red RESET button at the same time
11. Verify the red and green status lights both illuminate, indicating a successful reset
12. Release both buttons
13. To test the software reboot, push the “TEST” switch downwards until the green status light begins blinking. If the green status light is not blinking after 20 seconds, announce green light not blinking
14. Hold down the black ALT button while moving the CTRL switch to the “OFF” position
15. Close the cover on the CTRL switch
16. Verify the green status light has illuminated, indicating successful test
17. Announce Poor Comms RSSI error resolved

Comms Reboot Required (2441)

1. On the PDD, via the Power Override Panel for Subsystems, labeled “POPS”, move the Command and Data Handling switch, labeled “CDH” to the “OFF” position
2. Wait 5 seconds
3. Move the CDH switch to the “ON” position
4. Locate the antenna communications control component by identifying a square gray box with white turn knobs and a strip of indicator lights
5. If maximum RSSI strength is not indicated, proceed with the following steps
6. Locate the phase control knob, labeled “PHS”, rotate this until maximum possible RSSI improvement is achieved
7. Locate the modulation control knob, labeled “MOD”, rotate until maximum possible RSSI improvement is achieved
8. Locate the amplitude control knob, labeled “AMP”, rotate until maximum possible RSSI improvement is achieved
9. If the green indicator has not been lit, you may rotate these controls until maximum RSSI strength has been achieved, this is indicated by the green indicator being lit
10. On the SMC, verify the red status light has begun blinking
11. On SMC, press and hold the black ALT button while flipping the CTRL switch to the “ON” position
12. Verify the red status light is no longer illuminated
13. To finish resetting the software for the ACM, press and hold the black ALT button and the red RESET button at the same time
14. Verify both red and green status lights illuminate, indicating a successful reset
15. Release both buttons
16. To test the software reboot, push the “TEST” switch downwards until the green status light begins blinking. If the green status light is not blinking after 20 seconds, announce green light not blinking
17. Hold down the black ALT button while moving the CTRL switch to the “OFF” position
18. Close the cover on the CTRL switch
19. Verify the green status light has illuminated, indicating successful test
20. Announce Comms Reboot steps complete

Subsystem Power Bus Error (4968)

1. On the PDD, via POPS, move switches for “MPS”, “GNC”, “VSI”, and “CDH” to the “OFF” position
2. Check for any loose bus cables
3. If a cable is deemed loose, fully remove the cable from its socket
4. Firmly reinsert the cable into its socket
5. If no cable is deemed loose, fully remove and reinsert each cable into the respective sockets
6. Via POPS, move switches for “MPS”, “GNC”, “VSI”, and “CDH” to the “ON” position
7. Wait 5-10 seconds to verify the error clears before proceeding
8. Announce Subsystem Power Bus error resolved

Dust Sensor Error (2235)

1. Locate the dust sensor by identifying a cylindrical white housing smaller than the palm of a hand
2. Rotate the sensor body counterclockwise to unscrew it from its base
3. Once removed, set the sensor aside
4. Locate the replacement sensor and twist the protective cap counterclockwise to remove it, this cap may be put on the broken dust sensor
5. Insert the replacement sensor into the base and rotate clockwise until firmly secured
6. Announce Dust Sensor error resolved

Scientific Power Distribution Error (4280)

1. Locate the Scientific Management and Power Distribution (SMPD) component by identifying a gray rectangular housing labeled “SMPD2200xxxx”
2. Verify the SMPD indicator light is illuminated green, an unlit indicator implies an upstream power distribution issue and must be resolved first. Refer to the PDD display for which error is indicated
3. Locate the Scientific Power Regulation (SPR) component by identifying a small black housing labeled “SPR2200xxxx”
4. Take note of any illuminated red indicator lights on the SPR and their position
5. On the PDD, via POPS, move the Mission Payload Subsystem switch, labeled “MPS” to the “OFF” position
6. To prevent damage from unexpected electrical surges during operation, remove the power bus from its socket before proceeding
7. On the SPR, corresponding to the side that had a illuminated indicator, locate the fuse cartridge for that side and remove it
8. Prepare and insert a replacement fuse cartridge into the empty section
9. On the PDD, via POPS, reinsert the power bus
10. Move the MPS switch to the “ON” position
11. Verify the SMPD indicator is illuminated green and no indicators are illuminated on the SPR
12. Announce Small Fuse Box steps complete